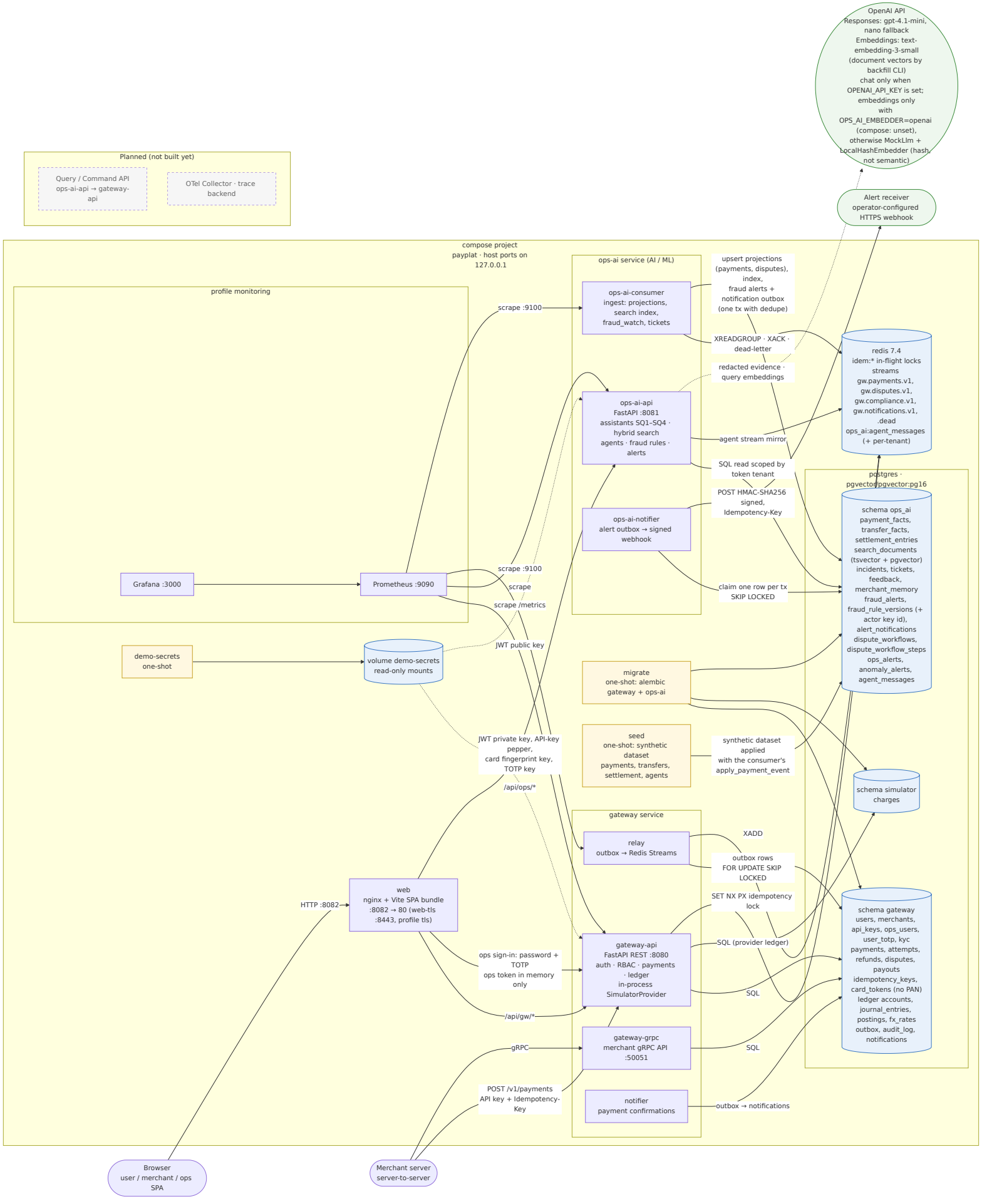

OpenAI API
Responses: gpt-4.1-mini, nano fallback
Embeddings: text-embedding-3-small (document vectors by backfill CLI)
chat only when OPENAI_API_KEY is set; embeddings only with OPS_AI_EMBEDDER=openai (compose: unset), otherwise MockLlm + LocalHashEmbedder (hash, not semantic)
Planned (not built yet)
Query / Command API
ops-ai-api → gateway-api
OTel Collector · trace backend
Alert receiver
operator-configured HTTPS webhook
compose project
payplat · host ports on 127.0.0.1
ops-ai service (AI / ML)
upsert projections (payments, disputes), index,
fraud alerts + notification outbox
(one tx with dedupe)
profile monitoring
scrape :9100
ops-ai-consumer
ingest: projections, search index, fraud_watch, tickets
XREADGROUP · XACK · dead-letter
redis 7.4
idem:* in-flight locks
streams
gw.payments.v1,
gw.disputes.v1,
gw.compliance.v1,
gw.notifications.v1,
.dead
ops_ai:agent_messages
(+ per-tenant)
redacted evidence · query embeddings
ops-ai-api
FastAPI :8081
assistants SQ1–SQ4 · hybrid search
agents · fraud rules · alerts
agent stream mirror
SQL read scoped by token tenant
postgres · pgvector/pgvector:pg16
POST HMAC-SHA256 signed, Idempotency-Key
ops-ai-notifier
alert outbox → signed webhook
schema ops_ai
payment_facts,
transfer_facts,
settlement_entries
search_documents
(tsvector + pgvector)
incidents, tickets,
feedback,
merchant_memory
fraud_alerts,
fraud_rule_versions (+ actor key id),
alert_notifications
dispute_workflows,
dispute_workflow_steps
ops_alerts,
anomaly_alerts,
agent_messages
Grafana :3000
Prometheus :9090
scrape :9100
scrape
scrape /metrics
claim one row per tx SKIP LOCKED
demo-secrets
one-shot
volume demo-secrets
read-only mounts
JWT public key
migrate
one-shot: alembic
gateway + ops-ai
seed
one-shot: synthetic dataset
payments, transfers, settlement, agents
synthetic dataset applied
with the consumer's apply_payment_event
JWT private key, API-key pepper,
card fingerprint key,
TOTP key
/api/ops/*
schema simulator
charges
gateway service
relay
outbox → Redis Streams
XADD
outbox rows
FOR UPDATE SKIP LOCKED
HTTP :8082
web
nginx + Vite SPA bundle
:8082 → 80 (web-tls
:8443, profile tls)
SET NX PX idempotency lock
schema gateway
users, merchants,
api_keys, ops_users,
user_totp, kyc
payments, attempts,
refunds, disputes,
payouts
idempotency_keys,
card_tokens (no PAN)
ledger accounts,
journal_entries,
postings, fx_rates
outbox, audit_log,
notifications
gateway-api
FastAPI REST :8080
auth · RBAC · payments · ledger
in-process SimulatorProvider
SQL (provider ledger)
SQL
ops sign-in: password + TOTP
ops token in memory only
/api/gw/*
gateway-grpc
merchant gRPC API
:50051
gRPC
SQL
POST /v1/payments
API key + Idempotency-Key
notifier
payment confirmations
outbox → notifications
Browser
user / merchant / ops SPA
Merchant server
server-to-server